

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****CERTIFICATE OF COMPLIANCE**

*This document is used to demonstrate compliance for nonresidential occupancies with requirements in §110.1, §110.3, §120.3, and §140.5, and with requirements in §141.0 for additions and alterations, for domestic water heating scopes using the prescriptive path. For high-rise residential and hotel/motel occupancies, compliance is demonstrated with requirements in §110.1, §110.3, §160.4 and §170.2(d), and with requirements in §180.1 for additions and §180.2 for alterations.*

|                               |                     |
|-------------------------------|---------------------|
| Project Name:                 | Enforcement Agency: |
| Dwelling Address:             | Permit Number:      |
| <b>A. GENERAL INFORMATION</b> |                     |

**A. GENERAL INFORMATION**

|                          |                                                         |                          |                                               |                          |               |                          |             |                          |                                |                          |                       |
|--------------------------|---------------------------------------------------------|--------------------------|-----------------------------------------------|--------------------------|---------------|--------------------------|-------------|--------------------------|--------------------------------|--------------------------|-----------------------|
| 01                       | Project Location (city)                                 |                          | 02                                            | Climate Zone             |               |                          |             |                          |                                |                          |                       |
| 03                       | Occupancy Types Within Project (select all that apply): |                          |                                               |                          |               |                          |             |                          |                                |                          |                       |
| <input type="checkbox"/> | Office                                                  | <input type="checkbox"/> | High-Rise Residential Multifamily > 4 stories | <input type="checkbox"/> | Relocatable   | <input type="checkbox"/> | School      | <input type="checkbox"/> | Restaurant/ Commercial Kitchen | <input type="checkbox"/> | Religious Facility    |
| <input type="checkbox"/> | State Building                                          | <input type="checkbox"/> | Healthcare Facility                           | <input type="checkbox"/> | Hotel/ Motel  | <input type="checkbox"/> | All Others  | <input type="checkbox"/> | Convention Center              | <input type="checkbox"/> | Medical Clinic        |
| <input type="checkbox"/> | Auditorium                                              | <input type="checkbox"/> | Parking Garage                                | <input type="checkbox"/> | Warehouse     | <input type="checkbox"/> | Retail      | <input type="checkbox"/> | Sports Arena                   | <input type="checkbox"/> | Gymnasium             |
| <input type="checkbox"/> | Classroom                                               | <input type="checkbox"/> | Library                                       | <input type="checkbox"/> | Theater       | <input type="checkbox"/> | Data Center | <input type="checkbox"/> | Support Areas                  | <input type="checkbox"/> | Financial Institution |
| <input type="checkbox"/> | Commercial/ Industrial                                  | <input type="checkbox"/> | School                                        | <input type="checkbox"/> | Grocery Store |                          |             |                          |                                |                          |                       |

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****B. PROJECT SCOPE**

*This table identifies any domestic water heating systems that are within the scope of the permit application and are demonstrating compliance using the prescriptive paths outlined in §140.5/ §170.2(d), and §141.0(a)/ §180.1, or §141.0(b)2N/ §180.2 for additions or alterations. Solar water heating systems should be documented on the NRCC-SAB compliance document. Combined hydronic water heating systems should be documented on the NRCC-MCH compliance document.*

| 01                                             |                                                                                          | 02                         | 03                                 |                                       |                                   |
|------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|---------------------------------------|-----------------------------------|
| My project consists of (check all that apply): |                                                                                          | System Type <sup>1,2</sup> | System Components                  |                                       |                                   |
| <input type="checkbox"/>                       | New System (DHW system being installed for the first time in newly constructed building) |                            | <input type="checkbox"/> Equipment | <input type="checkbox"/> Distribution | <input type="checkbox"/> Controls |
| <input type="checkbox"/>                       | System Alteration (equipment, distribution, or controls)                                 |                            | <input type="checkbox"/> Equipment | <input type="checkbox"/> Distribution | <input type="checkbox"/> Controls |

*Gas or propane central water heating systems serving multiple dwelling units require a solar water heating system per §170.2(d)3. Solar water heating systems are documented on the NRCC-SAB compliance document. If solar water heating systems are used in conjunction with non-solar systems to meet the water heating needs of the building, document the non-solar systems on this compliance document.*

*Note: If combined hydronic water heating systems are used in conjunction with domestic only systems to meet the water heating needs of the building, only document the domestic only systems on this compliance document.*

<sup>1</sup> FOOTNOTE: Point of use water heaters, or other non-central systems used to serve nonresidential spaces, are considered individual systems.

<sup>2</sup> Dwelling units refers to hotel/ motel guest rooms and units in a multifamily residential occupancy.

<sup>3</sup> DHW systems serving 2 or more dwelling units are considered "Central Systems" for multifamily and hotel/motel occupancies.

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****C. COMPLIANCE RESULTS**

*Table Instructions: Table C will indicate if the project data input into the compliance document is compliant with water heating requirements. This table is not editable by the user. If this table says "DOES NOT COMPLY" or "COMPLIES with Exceptional Conditions" refer to Table D., or the table indicated as not compliant for guidance.*

| 01                           | 02                   | 03       | 04       | 05                                                                |
|------------------------------|----------------------|----------|----------|-------------------------------------------------------------------|
| Domestic Hot Water Equipment | Distribution Systems | Controls | Pool/Spa | Compliance Results                                                |
| Table F                      | Table G              | Table H  | Table J  |                                                                   |
|                              |                      |          |          | COMPLIES, COMPLIES with Exceptional Conditions or DOES NOT COMPLY |

**D. EXCEPTIONAL CONDITIONS**

*This table is auto-filled with uneditable comments because of selections made or data entered in tables throughout the form.*

|  |
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|  |
|--|

**E. ADDITIONAL REMARKS**

*This table includes remarks made by the permit applicant to the Authority Having Jurisdiction.*

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|  |
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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****F. DOMESTIC HOT WATER EQUIPMENT**

*Table Instructions: Complete the following table to demonstrate compliance with mandatory equipment requirements in §110.1 and §110.3. Compliance with prescriptive requirements in §140.5(c)/§170.2(d) must also be demonstrated and with §141.0/§180.1/§180.2 for addition and alteration scopes.*

**Equipment Schedule: Water Heater Efficiency and Standby Loss**

|                  |          |               |                |                                     |                  |                             |                                          |                      |
|------------------|----------|---------------|----------------|-------------------------------------|------------------|-----------------------------|------------------------------------------|----------------------|
| 01               |          | 02            |                | 03                                  |                  |                             | 04                                       |                      |
| System Name:     |          | Space Type:   |                | Exception to §140.5(c)/§170.2(d)2B: |                  |                             | Capacity-weighted Average Efficiency (%) |                      |
| 05               | 06       | 07            | 08             | 09                                  | 10               | 11                          | 12                                       | 13                   |
| Name or Item Tag | Quantity | System Status | Equipment Type | Rated Input Capacity (Btu/h)        | Rated Efficiency | Minimum Efficiency Required | Efficiency Unit                          | Maximum Standby Loss |
|                  |          |               |                |                                     |                  |                             |                                          |                      |

<sup>1</sup> FOOTNOTE: In systems  $\geq 1\text{MMBtu/h}$  with multiple units, gas water heaters with input capacity  $> 100,000\text{ Btu/h}$  may meet the 90% Et requirement via an input capacity-weighted average.

<sup>2</sup> Compliant equipment may be found in the Modernized Appliance Efficiency Database System (MAEDBS) on the Energy Commission website: <https://cacertappliances.energy.ca.gov/Pages/Search/AdvancedSearch.aspx>

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****Multifamily Water Heating Equipment Serving Individual Dwelling Units- §170.2(d)1, §180.2(b)3C**

|                                         |                          |                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Equipment Type (select all that apply): |                          |                                                                                                                                                                                                                                                                                                                                                                 |
| System Name:                            |                          |                                                                                                                                                                                                                                                                                                                                                                 |
| 16                                      | <input type="checkbox"/> | Gas/propane instantaneous water heater with input rating $\leq$ 200,000 BTUH and no storage tank. Note: Cannot comply using the prescriptive path with a storage tank per Exception 1 §170.2(d)1 (New Construction and Additions Only)                                                                                                                          |
|                                         | <input type="checkbox"/> | A single 240-volt heat pump water heater serving the dwelling unit. (New Construction and Additions Only, §170.2(d)1A)                                                                                                                                                                                                                                          |
|                                         | <input type="checkbox"/> | A single 120-volt heat pump water heater serving a dwelling unit with 1 bedroom or less. (New Construction and Additions Only, Exception 2 to §170.2(d)1)                                                                                                                                                                                                       |
|                                         | <input type="checkbox"/> | A single heat pump water heater that meets the requirements of NEEA Advanced Water Heater Specification Tier 3 or higher (§170.2(d)1B or §180.2(b)3).                                                                                                                                                                                                           |
|                                         | <input type="checkbox"/> | A single heat pump water heater with storage tank located in the garage or conditioned space and be placed on an incompressible, rigid insulated surface with minimum R-10. The water heater shall be installed with a communication interface that meets either the requirements of 110.12(a) or has an ANSI/CTA-2045-B communication port. (Alterations Only) |
|                                         | <input type="checkbox"/> | If the existing water heater is an electric resistance water heater, a consumer electric water heater. (Alterations only, §180.2(b)3)                                                                                                                                                                                                                           |
| 17                                      | <input type="checkbox"/> | Replacement or altered gas or propane water heater (Alterations only, §180.2(b)3ci)                                                                                                                                                                                                                                                                             |

**Water Heating Equipment All Occupancies - §110.3(c)3, §140.5(a)**

|              | Yes                                                     | No                       | NA                       | Requirement                                                                                                                                                                                                                          |
|--------------|---------------------------------------------------------|--------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System Name: |                                                         |                          |                          |                                                                                                                                                                                                                                      |
| 18           | <input type="checkbox"/>                                | <input type="checkbox"/> | <input type="checkbox"/> | Unfired storage tank insulation shall have Internal + External $\geq$ R-16 OR External $\geq$ R-3.5. Label required per §110.3(c)3                                                                                                   |
| 19           | <input type="checkbox"/>                                | <input type="checkbox"/> | <input type="checkbox"/> | New state buildings 60% of energy for service water heating from site solar energy or recovered energy per §110.3(c)5                                                                                                                |
| 20           | <input type="checkbox"/>                                | <input type="checkbox"/> | <input type="checkbox"/> | Isolation valves for instantaneous water heater with input rating $>$ 6.8 kBTUH or 2 kW has been specified per §110.3(c)6                                                                                                            |
| 21           | <input type="checkbox"/>                                | <input type="checkbox"/> | <input type="checkbox"/> | School buildings $<$ 25,000ft <sup>2</sup> and $<$ 4 stories must install a heat pump water heating system per §140.5(a)1. Water heating systems serving an individual bathroom space may be an instantaneous electric water heater. |
| 22           | Air-Source Heat Pump Water Heaters (HPWHs) (§110.3(c)7) |                          |                          |                                                                                                                                                                                                                                      |
|              | <input type="checkbox"/>                                | <input type="checkbox"/> | <input type="checkbox"/> | Backup heat is required when inlet air is unconditioned unless the compressor cut-off temperature is below the Heating Winter Median of Extremes for the closest location listed in Table 2-3 from Reference Joint Appendix JA2.     |
|              | <input type="checkbox"/>                                | <input type="checkbox"/> | <input type="checkbox"/> | Consumer integrated HPWHs shall meet one of the following ventilation requirements:<br>Dropdown                                                                                                                                      |

**Multifamily and Hotel/Motel Water Heating Equipment for Central Systems Serving Dwelling Units §170.2(d)2&3**

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS**

|                                                                                                               |                          |                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Equipment Type (select all that apply):                                                                       |                          |                                                                                                                                                                                                                                                                                                                                                                             |
| System Name:                                                                                                  |                          |                                                                                                                                                                                                                                                                                                                                                                             |
| 23                                                                                                            | <input type="checkbox"/> | Heat pump Water Heater (New Construction and Additions Only- §170.2(d)2A)                                                                                                                                                                                                                                                                                                   |
|                                                                                                               |                          | <input type="checkbox"/> The primary heat pump water heater shall be a single-pass heat pump water heater. - §170.2(d)2Ai                                                                                                                                                                                                                                                   |
|                                                                                                               |                          | <input type="checkbox"/> The primary storage tank temperature setpoint shall be at least 135°F. - §170.2(d)2Aiv                                                                                                                                                                                                                                                             |
|                                                                                                               |                          | <input type="checkbox"/> The minimum heat pump water heater compressor cut-off temperature shall be equal to or lower than 40°F ambient air temperature. - §170.2(d)2Avi                                                                                                                                                                                                    |
|                                                                                                               |                          | <input type="checkbox"/> Design documentation shall specify the operating conditions at which the primary heat pump water heater can supply hot water at design outlet water temperature without engaging auxiliary heating mechanism. - §170.2(d)2Avii                                                                                                                     |
| 24                                                                                                            | <input type="checkbox"/> | NEEA Advanced Water Heater Specification for commercial heat pump water heater Tier 2 or higher (§170.2(d)2A)                                                                                                                                                                                                                                                               |
| 25                                                                                                            | <input type="checkbox"/> | Gas/propane water heater (New Construction and Additions Only, §170.2(d)2B)                                                                                                                                                                                                                                                                                                 |
|                                                                                                               |                          | <input type="checkbox"/> A solar water-heating system meeting the installation criteria specified in Reference Residential Appendix RA4 and with a minimum solar savings fraction of 0.20 in Climate Zones 1 through 9 or a minimum solar savings fraction of 0.35 in Climate Zones 10 through 16 - §170.2(d)2Bi. This system should be documented on the NRCC-SAB form OR; |
|                                                                                                               |                          | <input type="checkbox"/> A solar water-heating system meeting the installation criteria specified in Reference Residential Appendix RA4 and with a minimum solar savings fraction of 0.15 in Climate Zones 1 through 9 or a minimum solar savings fraction of 0.30 in Climate Zones 10 through 16. - §170.2(d)2Bii. This system should be documented on the NRCC-SAB form.  |
| AND a drain water heat recovery system that is field verified as specified in the Reference Appendix RA3.6.9. |                          |                                                                                                                                                                                                                                                                                                                                                                             |

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****G. DOMESTIC HOT WATER DISTRIBUTION SYSTEM**

*Table Instructions: Complete the following table to demonstrate compliance for nonresidential occupancies with distribution requirements in §120.3 and §140.5. For multifamily and hotel/motel occupancies, compliance is demonstrated with requirements in §110.3(c), §160.4 and §170.2(d).*

*Recirculation Loops in Central Systems Serving Dwelling Units, Hotel/Motel Guest Rooms or Nonresidential Spaces - §110.3, §170.2(d)2*

|    | Yes                      | No                       | Requirement                                                                                                                                                                                                                                                                              |
|----|--------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | <input type="checkbox"/> | <input type="checkbox"/> | Air release valve or vertical pump installation per §110.3(c)4A                                                                                                                                                                                                                          |
| 02 | <input type="checkbox"/> | <input type="checkbox"/> | Check valve or similar located between recirculation pump and water heating equipment to prevent backflow per §110.3(c)4B                                                                                                                                                                |
| 03 | <input type="checkbox"/> | <input type="checkbox"/> | Hose bibb installed between pump and equipment and isolation valve between hose bibb and equipment per §110.3(c)4C                                                                                                                                                                       |
| 04 | <input type="checkbox"/> | <input type="checkbox"/> | Isolation valves on both sides of the pump per §110.3(c)4D                                                                                                                                                                                                                               |
| 05 | <input type="checkbox"/> | <input type="checkbox"/> | Cold water and recirculation loop piping shall not be connected to the hot water storage tank drain port per §110.3(c)4E                                                                                                                                                                 |
| 06 | <input type="checkbox"/> | <input type="checkbox"/> | Check valve installed on cold water supply between hot water system and next closest tee on cold water supply per §110.3(c)4F                                                                                                                                                            |
| 07 | <input type="checkbox"/> | <input type="checkbox"/> | DWELLING UNITS ONLY: For heat pump water heating systems, the hot water return from the recirculation loop shall connect to a recirculation loop tank and shall not directly connect to the primary heat pump water heater inlet or the primary thermal storage tanks per §170.2(d)2Aii. |
| 08 | <input type="checkbox"/> | <input type="checkbox"/> | DWELLING UNITS ONLY: For heat pump water heating systems, the fuel source for the recirculation loop tank shall be electricity if per §170.2(d)2Aiii.                                                                                                                                    |
| 09 | <input type="checkbox"/> | <input type="checkbox"/> | DWELLING UNITS ONLY: The recirculation loop tank temperature setpoint shall be at least 10°F lower than the primary thermal storage tank temperature setpoint - §170.2(d)2Av                                                                                                             |
| 10 | <input type="checkbox"/> | <input type="checkbox"/> | DWELLING UNITS ONLY: All hot water distribution piping shall be sized in accordance with the California Plumbing Code Appendix M per §170.2(d)2C.                                                                                                                                        |
| 11 | <input type="checkbox"/> | <input type="checkbox"/> | DWELLING UNITS ONLY: A recirculation system with mechanical or digital thermostatic master mixing valve on each distribution supply and return loop and meet the requirements in Reference Appendix RA4.4, unless building has ≤ 8 dwelling units, per §170.2(d)2D.                      |
| 12 | <input type="checkbox"/> | <input type="checkbox"/> | DWELLING UNITS ONLY: Insulation for hot water pipes shall be field verified as specified in Reference Appendix RA3.6.3 per §170.2(d)2E.                                                                                                                                                  |

**Distribution of Individual System(s) serving Dwelling Units - §170.2(d)1**



## DOMESTIC WATER HEATING SYSTEM

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS**

|              | Yes                      | Not Applicable           | Requirement                                                                                                                                                                                                         |                                                                                                                              |
|--------------|--------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| System Name: |                          |                          |                                                                                                                                                                                                                     |                                                                                                                              |
| 13           | <input type="checkbox"/> | <input type="checkbox"/> | Per §170.2(d)1A, single 240-volt heat pump water heaters serving dwelling units must also include systems with:                                                                                                     |                                                                                                                              |
|              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>                                                                                                                                                                                            | Compact hot water distribution system as specified in Reference Appendix RA4.4.16 in climate zone 1 & 16; AND                |
|              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>                                                                                                                                                                                            | A drain water heat recovery system that is field verified by a HERS Rater per Reference Appendix RA3.6.9 in climate zone 16. |
| 14           | <input type="checkbox"/> | <input type="checkbox"/> | A drain water heat recovery system that is field verified by a HERS Rater per Reference Appendix RA3.6.9 in climate zone 16 per §170.2(d)1B.                                                                        |                                                                                                                              |
| 15           | <input type="checkbox"/> | <input type="checkbox"/> | For recirculation distribution systems serving individual dwelling units, only Demand Recirculation Systems with manual on/off control as specified in the Reference Appendix RA4.4.9 shall be used per §170.2(d)1. |                                                                                                                              |



**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****Mandatory Pipe Insulation All Occupancies - §120.3, §160.4**

|              | Yes                      | Not Applicable           | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|--------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System Name: |                          |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 16           | <input type="checkbox"/> | <input type="checkbox"/> | <p>For systems serving dwelling units and common areas, pipe insulation must meet the minimum insulation requirements in Table 160.4-A (see below) except:</p> <ul style="list-style-type: none"> <li>- Piping that penetrates framing members shall not be required to have pipe insulation for the distance of the framing penetration. Piping that penetrates metal framing shall use grommets, plugs, wrapping or other insulating material to assure that no contact is made with the metal framing. Insulation shall abut securely against all framing members.</li> <li>- Piping installed in interior or exterior walls shall not be required to have pipe insulation if all of the requirements are met for compliance with Quality Insulation Installation (QII) as specified in the Reference Residential Appendix RA3.5.</li> <li>- Piping surrounded with a minimum of 1 inch of wall insulation, 2 inches of crawlspace insulation, or 4 inches of attic insulation, shall not be required to have pipe insulation.</li> </ul> |
| 17           | <input type="checkbox"/> | <input type="checkbox"/> | <p>For systems serving nonresidential spaces, pipe insulation for the following applications is specified to comply with Table 120.3-A1 or Table 120.3-A2 (see below) per §120.3:</p> <ul style="list-style-type: none"> <li>- Recirculating system piping, including supply and return piping of the water heater</li> <li>- The first 8 ft of hot and cold outlet piping, including between storage tank and heat trap, for a nonrecirculating storage system</li> <li>- Pipes that are externally heated</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 18           | <input type="checkbox"/> | <input type="checkbox"/> | <p>Per §120.3(b)/ §160.4(f), pipe and appurtenance insulation shall be protected from damage, including that due to sunlight, moisture, equipment maintenance, and wind. Protection shall include:</p> <ul style="list-style-type: none"> <li>• Insulation exposed to weather shall be installed with a cover suitable for outdoor service.</li> <li>• Pipe insulation buried below grade must be installed in a water-proof and non-crushable casing or sleeve.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**TABLE 160.4-A PIPE INSULATION THICKNESS**

| Fluid Temperature Range (°F) | Conductivity Range (Btu-in per hour per ft <sup>2</sup> per °F) | Insulation Mean Rating Temp (°F) | Nominal Pipe Diameter (in)  |                  |                |                                      |
|------------------------------|-----------------------------------------------------------------|----------------------------------|-----------------------------|------------------|----------------|--------------------------------------|
|                              |                                                                 |                                  | < 1                         | 1 to < 1.5       | 1.5 to < 4     | 1.5 to < 4 Multifamily & Hotel/Motel |
|                              |                                                                 |                                  | Minimum Insulation Required |                  |                |                                      |
| 105-140                      | 0.22-0.28                                                       | 100                              | 1.0 in or R-7.7             | 1.5 in or R-12.5 | 1.5 in or R-11 | 2 in or R-16                         |

**H. DOMESTIC HOT WATER SYSTEM CONTROLS - §110.3, §120.9, §160.4(d), §170.2(d), §180.1(b)3**

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS**

*Table Instructions: Complete the following table to demonstrate compliance with controls requirements in §110.3 for all occupancies. For nonresidential occupancies, compliance is demonstrated with requirements in §120.9. For multifamily residential and hotel/motel occupancies, compliance is demonstrated with requirements in §160.4(d) and §170.2(d).*

| System Name: |                          |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------|--------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | Yes                      | Not Applicable           | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 01           | <input type="checkbox"/> | <input type="checkbox"/> | Construction documents require manufacturer certification that service water-heating systems are equipped with automatic temperature controls capable of adjusting temperature settings per §110.3(a).                                                                                                                                                                                                                                                                   |
| 02           | <input type="checkbox"/> | <input type="checkbox"/> | Systems with capacity > 167,000 BTUH equipped with outlet temperature controls per §110.3(c)1 unless covered by California Plumbing Code Section 613.0.                                                                                                                                                                                                                                                                                                                  |
| 03           | <input type="checkbox"/> | <input type="checkbox"/> | Controls for circulating pumps or electrical heat trace systems are capable of automatically turning off the system per §110.3(c)2 unless system serves healthcare facility.                                                                                                                                                                                                                                                                                             |
| 04           | <input type="checkbox"/> | <input type="checkbox"/> | For recirculation systems serving multiple dwelling units, design includes a mechanical or digital thermostatic master mixing valve on each distribution supply and return loop per §170.2(d), or §180.1(b)3 for additions.                                                                                                                                                                                                                                              |
| 05           | <input type="checkbox"/> | <input type="checkbox"/> | For recirculation systems serving individual dwelling units, design includes manual on/off controls as specified in Reference Appendix RA 4.4.9 per §170.2(d).                                                                                                                                                                                                                                                                                                           |
| 06           | <input type="checkbox"/> | <input type="checkbox"/> | Combustion air positive shut-off shall be provided per §120.9(a) & §160.4(d) on all newly installed commercial boilers as follows:<br><ul style="list-style-type: none"> <li>- Boiler with input capacity <math>\geq</math> 2.5 MMBtu/h, in which the boiler is designed to operate with a nonpositive vent static pressure</li> <li>- Boilers where one stack serves two or more boilers with a total combined input capacity per stack of 2.5 MMBtu/h.</li> </ul>      |
| 07           | <input type="checkbox"/> | <input type="checkbox"/> | Boiler combustion air fans with motors $\geq$ 10 hp shall meet one of the following for newly installed boilers per §120.9(b) & §160.4(d)2:<br><ul style="list-style-type: none"> <li>- The fan motor shall be driven by a variable speed drive OR</li> <li>- The fan motor shall include controls that limit the fan motor demand to <math>\leq</math> 30% of the total design wattage at 50% of the design air volume.</li> </ul>                                      |
| 08           | <input type="checkbox"/> | <input type="checkbox"/> | Newly installed boilers with an input capacity $\geq$ 5 MMBtu/h and a steady state full-load combustion efficiency < 90% shall maintain excess (stack-gas) oxygen concentrations $\leq$ 5% by volume on a dry basis over firing rates of 20-100%. Combustion air volume shall be controlled with respect to firing rate or flue gas oxygen concentration. Use of a common gas and combustion air control linkage or jack shaft is prohibited per §120.9(c) & §160.4(d)3. |

**I. ELECTRIC READY BUILDINGS**

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS**

*This table includes electric ready requirements that must be met when installing gas or propane water heaters in dwelling units to demonstrate compliance with §160.9.*

|    |                                                                        |                                                                                     |                                                                                   |                                        |
|----|------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------|
| 01 | Systems serving multifamily occupancy that use gas or propane include: | <input type="checkbox"/> Gas/propane water heater serving individual dwelling units | <input type="checkbox"/> Gas/propane water heater serving multiple dwelling units | <input type="checkbox"/> None of these |
|----|------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------|

**Gas/Propane Water Heaters Serving Individual Dwelling Units - §160.9(e)**

|              |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System Name: |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|              | Yes                      | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 02           | <input type="checkbox"/> | 160.9(e)2: A condensate drain that is no more than 2 inches higher than the base of the installed water heater, and allows natural draining without pump assistance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 03           | <input type="checkbox"/> | <p>A ventilation method meeting the following per 160.9(e)4:<br/> Dropdown options (only the one that is selected shows up):</p> <ul style="list-style-type: none"> <li>- The designated space for the future heat pump water heater shall have a minimum volume of 700 cubic feet; or</li> <li>- The designated space for the future heat pump water heater shall vent to a communicating space in the same pressure boundary via permanent openings with a minimum total NFA of 250 square inches., so that the total combined volume connected via permanent openings is 700 cubic feet or larger. The permanent openings shall be: <ul style="list-style-type: none"> <li>- Fully louvered doors with fixed louvers; or</li> <li>- Two permanent fixed openings located within 12 inches from the enclosure top and bottom;</li> </ul> </li> <li>- The designated space for the future heat pump water heater shall include two 8 inches capped ducts, venting to the building exterior: <ul style="list-style-type: none"> <li>- All ducts, connections, and building penetrations shall be sealed.</li> <li>- Exhaust air ducts and all ducts which cross pressure boundaries shall be insulated to a minimum insulation level of R-6.</li> <li>- Airflow from termination points shall be diverted away from each other.</li> </ul> </li> </ul> |

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****Gas/Propane Water Heaters Serving Multiple Dwelling Units - §160.9(f)**

|              |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System Name: |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|              | Yes                      | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 04           | <input type="checkbox"/> | <p>160.9(f)2A: Space shall be reserved for a Heat Pump. The minimum space reserved shall include space for service clearances and air flow clearances and shall meet the following:<br/> <i>Dropdown options (only the one that is selected shows up):</i></p> <ul style="list-style-type: none"> <li>- The space reserved shall be the space required for a heat pump water heater system that meets the total building hot water demand as calculated and documented by the responsible person associated with the project; or</li> <li>- The space reserved shall meet the requirements specified in Joint Appendix JA15.2.1.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 05           | <input type="checkbox"/> | <p>160.9(f)2B: Space shall be reserved for Tanks. The minimum space reserved shall include space for service clearances and shall meet the following:<br/> <i>Dropdown options (only the one that is selected shows up):</i></p> <ul style="list-style-type: none"> <li>- The space reserved shall be the space required for a heat pump water heater system that meets the total building hot water demand as calculated and documented by the responsible person associated with the project; or</li> <li>- The space reserved shall meet the requirements specified in Joint Appendix JA15.2.2.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 06           | <input type="checkbox"/> | <p>160.9(f)3: Ventilation shall be provided by meeting the following:<br/> <i>Dropdown options (only the one that is selected shows up):</i></p> <ul style="list-style-type: none"> <li>- Physical space reserved for the heat pump shall be located outside; or</li> <li>- A pathway shall be reserved for future routing of supply and exhaust air via ductwork from the reserved heat pump location to a suitable outdoor location. Penetrations through the building envelope for louvers and ducts shall be planned and identified for future use. The reserved pathway and penetrations through the building envelope shall be sized to meet one of the following: <ul style="list-style-type: none"> <li>- The reserved pathway and penetrations shall be sized to serve a heat pump water heater system that meets the total building hot water demand as calculated and documented by the responsible person associated with the project.</li> <li>- The reserved pathway and penetrations shall be sized to meet the requirements specified in Joint Appendix JA15.2.3.</li> </ul> </li> </ul> |
| 07           | <input type="checkbox"/> | <p>160.9(f)4: Condensate drainage piping. An approved receptacle that is sized per the California Plumbing Code for condensate drainage shall be installed within 3 feet of the reserved heat pump location, or piping shall be installed from within 3 feet of the reserved heat pump location to an approved discharge location that is sized in accordance with the California Plumbing Code, and meet the following:<br/> <i>Dropdown options (only the one that is selected shows up):</i></p> <ul style="list-style-type: none"> <li>- Condensate drainage shall be sized to serve a heat pump water heater system that meets the total building hot water demand as calculated and documented by the responsible person associated with the project.</li> <li>- Condensate drainage piping shall be sized to meet the requirements specified in Joint Appendix JA15.2.4.</li> </ul>                                                                                                                                                                                                               |

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****J. POOL & SPAs**

*This table documents compliance with mandatory pool/spa requirements in §110.4/§160.7.*

| Is the pool/spa heated? | Heated/Not Heated     |            |                |              |        |                   |                       |        |                                        |
|-------------------------|-----------------------|------------|----------------|--------------|--------|-------------------|-----------------------|--------|----------------------------------------|
| 01                      | 02                    | 03         | 04             | 05           | 06     | 07                | 08                    | 09     | 10                                     |
| Pool/ Spa Description   | Pool/Spa Service Type | Efficiency | On/Off Control | Instructions | Covers | Heating Equipment | Heating Source Sizing | Piping | Pool Directional Inlets & Pump Control |
|                         |                       |            |                |              |        |                   |                       |        |                                        |

**K. DECLARATION OF REQUIRED CERTIFICATES OF INSTALLATION****Low-rise Multifamily and Multifamily Mixed-use Certificates of Installation**

*Table Instructions: Selections have been made based on information provided in previous tables of this document. If any selection needs to be changed, please explain why in Table E. Additional Remarks. These documents must be provided to the building inspector during construction and can be found online.*

| Yes                              | No                    | Form/Title                                          | Field Inspector          |                          |
|----------------------------------|-----------------------|-----------------------------------------------------|--------------------------|--------------------------|
|                                  |                       |                                                     | Pass                     | Fail                     |
| <input checked="" type="radio"/> | <input type="radio"/> | NRCI-PLB-01-E - Must be submitted for all buildings | <input type="checkbox"/> | <input type="checkbox"/> |

**L. DECLARATION OF REQUIRED CERTIFICATES OF ACCEPTANCE**

*There are no Certificates of Acceptance Applicable to service water heating requirements.*

**M. DECLARATION OF REQUIRED CERTIFICATES OF VERIFICATION**

***Nonresidential, hotel/motel, and high-rise multifamily and multifamily mixed-use certificates of verification.***

*There are no certificates of verification for low-rise multifamily and mixed-use buildings complying prescriptively.*

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****DOCUMENTATION AUTHOR'S DECLARATION STATEMENT**

1. I certify that this Certificate of Compliance documentation is accurate and complete.

|                            |                                                           |
|----------------------------|-----------------------------------------------------------|
| Documentation Author Name: | Documentation Author Signature:                           |
| Company:                   | Signature Date:                                           |
| Address:                   | CEA/AEA/ECC Certification Identification (if applicable): |
| City/State/Zip:            | Phone:                                                    |

**RESPONSIBLE PERSON'S DECLARATION STATEMENT**

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Compliance is true and correct.
2. I am eligible under Division 3 of the Business and Professions Code to accept responsibility for the building design or system design identified on this Certificate of Compliance (responsible designer).
3. The energy features and performance specifications, materials, components, and manufactured devices for the building design or system design identified on this Certificate of Compliance conform to the requirements of Title 24, Part 1 and Part 6 of the California Code of Regulations.
4. The building design features or system design features identified on this Certificate of Compliance are consistent with the information provided on other applicable compliance documents, worksheets, calculations, plans and specifications submitted to the enforcement agency for approval with this building permit application.
5. I understand that a completed signed copy of this Certificate of Compliance shall be made available with the building permit(s) issued for the building and shall be made available to the enforcement agency for all applicable inspections. I will take the necessary steps to fulfill this requirement.
6. I understand that a completed signed copy of this Certificate of Compliance is required to be included with the documentation the builder provides to the building owner at occupancy. I will take the necessary steps to fulfill this requirement.

|                            |                                 |
|----------------------------|---------------------------------|
| Responsible Designer Name: | Responsible Designer Signature: |
| Company :                  | Date Signed:                    |
| Responsible Person Scope:  |                                 |
| Address:                   | License:                        |
| City/State/Zip:            | Phone:                          |

|                                               |               |
|-----------------------------------------------|---------------|
| CERTIFICATE OF COMPLIANCE – USER INSTRUCTIONS | NRCC-PLB-01-E |
| Domestic Water Heating System                 | (Page 1 of 3) |

#### A. General Information

1. Enter the City the project is located in.
2. Climate Zone: Select from dropdown.
3. Select the applicable Occupancy Types within the project.

#### B. Project Scope

1. Select whether the project is New or Altered.
2. System Type: Select from dropdown.
3. Select the System Components.

#### C. Compliance Results

1. Results in this table are automatically calculated from data input and calculations in Tables F through H.

#### D. Exceptional Conditions

1. This table is auto-filled with uneditable comments because of selections made or data entered in tables throughout the form.

#### E. Additional Remarks

1. Enter any notes or comments for the AHJ.

#### F. Domestic Hot Water Equipment

##### Equipment Schedule: Central Systems

3. Enter the System Name.
4. Exception to §140.5(c)/§170.2(d)3: Select from dropdown.
5. Check the box if the Capacity-weight efficiencies for gas systems  $\geq 1\text{MMBtu/h1}$
6. This field is filled out automatically.
7. Enter the Name or Item Tag.
8. Equipment Type: Select from dropdown.
9. Enter the Volume.
10. Enter the Rated Input Capacity.
11. Enter the Rated Efficiency.
12. This field is filled out automatically.
13. This field is filled out automatically
14. Enter the Designed Standby Loss.
15. This field is filled out automatically

|                                               |               |
|-----------------------------------------------|---------------|
| CERTIFICATE OF COMPLIANCE – USER INSTRUCTIONS | NRCC-PLB-01-E |
| Domestic Water Heating System                 | (Page 2 of 3) |

### **Water Heating Equipment in Individual Dwelling Units**

16. Select the applicable equipment within the project.

17. Select if the project meets the listed requirements.

### **Water Heating Equipment All Occupancies**

18–21. Select if the project meets the listed requirements.

### **Water Heating Equipment for Central Systems Serving Dwelling Units §170.2(d)2&3**

22. Select the applicable equipment within the project.

## **G. Domestic Hot Water Distribution System**

### **Recirculation Loops in Central Systems Serving Dwelling Units or Nonresidential Spaces**

1-9. Select if the project meets the listed requirements.

Distribution of Individual System(s) serving Dwelling Units

10-12. Select if the project meets the listed requirements.

Mandatory Pipe Insulation All Occupancies

13-15. Select if the project meets the listed requirements.

## **H. Domestic Hot Water System Controls**

1-8. Select if the project meets the listed requirements.

## **I. Declaration of Required Certificates of Installation**

Selections have been automatically made based on information provided in this document. If any selections have been changed by the permit applicant, an explanation should be included in Table E. Additional Remarks.

## **J. Declaration of Required Certificates of Acceptance**

Selections have been made based on information provided in this document. If any selections have been changed by the permit applicant, an explanation should be included in Table E. Additional Remarks.



|                                               |               |
|-----------------------------------------------|---------------|
| CERTIFICATE OF COMPLIANCE – USER INSTRUCTIONS | NRCC-PLB-01-E |
| Domestic Water Heating System                 | (Page 3 of 3) |

#### Documentation Declaration Statements

1. The person who prepared the NRCC will sign and complete the fields for their name, company (if applicable), address, phone number, certification information (if applicable), date and signature.
2. The person who is assuming responsibility for the project being built to comply with Title 24, Part 6, will complete the fields for their name, company (if applicable), address, phone number, license number (if applicable), date and signature.